

Srinivasa Ajay Babu

AI & Machine Learning Engineer

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PROFESSIONAL SUMMARY

Generative AI & Machine Learning Engineer with 3+ years of progressive, project-based experience building Generative AI, Artificial Intelligence, and Machine Learning solutions with Python across healthcare and banking/financial services platforms. Skilled in designing LLM-powered applications, fine-tuning and prompting large language models, building RAG pipelines, and developing Python-based ML models for classification, fraud detection, and automation. Hands-on experience delivering enterprise GenAI/AI/ML solutions for Optum, Wells Fargo, and Olive Crypto Solutions, with strong grounding in context engineering, data pipelines, A/B testing, and model evaluation to drive measurable business impact. Proven ability to collaborate cross-functionally and translate complex AI/ML capabilities into production-ready outcomes.

TECHNICAL SKILLS

- **Programming & ML:** Python, Pandas, NumPy, Scikit-learn, ML Model Development & Evaluation, Feature Engineering, SQL, REST APIs, JSON, ETL Pipelines
- **Generative AI:** OpenAI GPT-4/3.5, Gemini API, Prompt Engineering, LangChain, RAG, LLM Fine-Tuning, Hugging Face, Transformers, Tokenization, Prompt Templates, Few-shot & Zero-shot Learning, LLM Content Generation, Guidelines-Based Classification
- **Artificial Intelligence:** Conversational AI / Chatbots, NLP, Knowledge Graphs, Semantic Search, Embeddings, Vector Databases (Pinecone, FAISS, ChromaDB), Human Evaluation Methodologies, AI System Design
- **Context Engineering & Grounding:** URL Crawling, SERP Analysis, Data Pipelines for Prompts, A/B Testing, Experiment Design, Prompt Scoring Metrics
- **Cloud & DevOps:** AWS (S3, Lambda, EC2, Bedrock), Azure OpenAI, GCP, Docker, Git, CI/CD Pipelines
- **Data & Storage:** Snowflake, PostgreSQL, MySQL, ETL Pipelines
- **Visualization & Analytics:** Power BI, Tableau, Excel

WORK EXPERIENCE

- **Optum** Aug 2025 - Present
AI & Machine Learning Engineer
 - **Generative AI Solutions:** Designed and fine-tuned GPT-4 and Gemini API solutions for clinical documentation, member support automation, and healthcare guidelines-based classification.
 - **Machine Learning:** Built and deployed Python models with Scikit-learn, Pandas, and NumPy for risk stratification and care-gap prediction using healthcare and claims data.
 - **AI Assistants:** Prototyped AI-powered virtual assistants and workflow copilots to streamline member and provider support interactions.
 - **RAG & Accuracy:** Implemented LangChain and Pinecone RAG pipelines grounded in clinical guidelines and policy documents, improving response accuracy by 38%.
 - **Evaluation:** Executed A/B testing and prompt performance analysis, translating quantitative and qualitative findings into measurable prompt and model improvements.
 - **APIs & Integration:** Built scalable Python FastAPI services for GenAI/ML orchestration and integration with downstream healthcare systems.
 - **Hallucination Reduction:** Reduced hallucinations by 35% using few-shot prompting, temperature tuning, and system instruction optimization.
 - **Compliance:** Ensured HIPAA-compliant handling of PHI/PII across all GenAI and ML pipelines.
 - **Environment:** Python, OpenAI GPT-4, Gemini API, LangChain, Scikit-learn, Pandas, Pinecone, AWS (Lambda, EC2, S3), FastAPI, Docker, Git, REST APIs
- **Wells Fargo** Jan 2025 - Jul 2025
AI & Machine Learning Engineer
 - **AI/ML Solutions:** Designed Generative AI and Machine Learning solutions in Python for fraud detection, transaction analysis, and customer support automation in a regulated banking environment.
 - **Banking Assistant:** Developed an Azure OpenAI and GPT-powered assistant for customer query automation, improving response accuracy by 45%.
 - **Security & Governance:** Built compliant prompt and ML frameworks for financial data, including PII masking and governance controls aligned to banking regulations.
 - **Grounded Responses:** Implemented RAG architecture using Azure Cognitive Search and FAISS for grounded, policy-compliant responses.
 - **Fraud Risk:** Built Python ML models for transaction anomaly and fraud-risk scoring, integrated into real-time banking workflows.
 - **Agentic Workflows:** Designed conversational AI flows using LangChain agents and tool integrations for banking operations support.
 - **Data & Analytics:** Used SQL and Snowflake for structured retrieval in AI/ML pipelines and built Power BI dashboards to monitor model and prompt KPIs.
 - **Environment:** Python, Azure OpenAI, LangChain, FAISS, Scikit-learn, SQL, Snowflake, Power BI, Docker, Git, REST APIs

• Olive Crypto Solutions

Jun 2022 - Dec 2023

AI & Machine Learning Engineer

- **Generative AI:** Designed AI solutions for an India-based banking client covering customer onboarding, KYC/AML support, and multilingual customer service automation.
- **Risk & Fraud Models:** Built Python ML models for credit risk scoring, transaction fraud detection, and UPI-based payment anomaly detection.
- **Chatbots & Assistants:** Developed OpenAI GPT and NLP-powered chatbots and workflow assistants for retail and digital banking customers across India.
- **Regulatory Compliance:** Ensured compliance with RBI data localization and KYC/AML guidelines throughout AI/ML data handling and deployment.
- **Semantic Search:** Implemented embeddings and ChromaDB to power policy and product-guideline lookup for customer support teams.
- **RAG Pipelines:** Retrieved banking product guidelines, regulatory policies, and customer transaction history for grounded AI responses.
- **Model Evaluation:** Improved AI recommendation accuracy by 30% through prompt tuning, contextual grounding, and Python-based evaluation.
- **NLP & APIs:** Used SpaCy and NLTK for preprocessing and entity recognition; developed Flask/FastAPI services and automated data workflows with ETL pipelines.
- **Environment:** Python, OpenAI GPT, LangChain, ChromaDB, Scikit-learn, SQL, SpaCy, NLTK, Flask, FastAPI, AWS

• Olive Crypto Solutions

Jan 2022 - May 2022

Python Developer Intern

- **Data Pipelines:** Assisted in building and testing Python scripts and data pipelines for banking data ingestion, cleaning, and transformation workflows.
- **Data Validation:** Wrote SQL queries and Python Pandas/NumPy utilities to validate and preprocess structured banking datasets for downstream ML use.
- **Testing & Debugging:** Supported senior engineers in unit testing and debugging Python modules used in AI/ML data pipelines.
- **Foundational Experience:** Gained hands-on exposure to Machine Learning fundamentals and REST API integration under the guidance of the AI engineering team.
- **Environment:** Python, Pandas, NumPy, SQL, Git, REST APIs

EDUCATION

• University of Southern Mississippi, USA

Master's in Computer Science

- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Natural Language Processing, Deep Learning, Data Structures & Algorithms, Cloud Computing

• Lakki Reddy Bali Reddy Engineering College

Bachelor's in Engineering

- **Relevant Coursework:** Python Programming, Database Management Systems, Data Structures, Software Engineering, Data Analytics

CERTIFICATIONS

- **AWS Certified Cloud Practitioner**
- **Microsoft Azure AI Engineer Associate**
- **Prompt Engineering for Generative AI** (Coursera / DeepLearning.AI)

KEY PROJECTS & ACHIEVEMENTS

- **Clinical GenAI Assistant - Optum:** Built a LangChain and Pinecone RAG assistant grounding clinical documentation in guidelines; reduced hallucinations by 35% and improved response accuracy.
- **Banking Virtual Assistant - Wells Fargo:** Developed an Azure OpenAI-powered banking assistant for fraud detection and customer support automation, improving response accuracy.
- **Credit Risk & UPI Fraud Detection - Olive Crypto Solutions:** Built Python ML models for credit risk scoring and UPI payment anomaly detection, improving AI recommendation accuracy.
- **Multi-step Prompt Chaining & RAG Pipelines:** Designed reusable Pinecone, FAISS, and ChromaDB RAG and prompt-chaining frameworks reused across healthcare and banking projects to ground LLM responses in enterprise data.
- **Certifications While Delivering Production Systems:** Earned AWS Certified Cloud Practitioner and Microsoft Azure AI Engineer Associate while actively building and shipping GenAI/ML solutions.